i) Definition: CO1- Assessment. P. Prakash
192524162

cloud computing is a model that enables network access to a shared pool of configurable computing resources such as server, storage, network application and services. Instead of purchasing and manufacturing physical hardware users can access computing resources over the internet and pay only for the service they use. This pay per-use model makes cloud computing cost effective, flexible and highly scalable.

characteristics:

=> On demand self-service:-

In traditional systems, purchasing and configuring hardware may take several days or weeks whereas cloud resources are available within minutes.

=> Broad Network Access:-

Traditional on-premise system are generally limited to the organization internal network whereas cloud services support access from any location.

2) The evolution of computer hardware laid the foundation for cloud computing by continuously improving processing power, storage capacity and networking capabilities.

Hardware Evolution.

=> 1939, Berry brothers invented vaccum tubes } 1st gen

=> 1946 ENIAC General purpose computer } 2nd gen

=> 1947 IC used in minicomputer, intel 4004 } 3rd gen

=> 1971 commercially available personal computer } 4th

Internet software Evolution:-

=> 1957, Launch of sputnik I led US establish A Advanced research project Agency.

=> SAGE semi-Automatic Ground Environment and with 800 programmers.

=> IPv6 (1995)

=> 1st web server by Jim-Berners Lee

=> Mosaic Browser

=> TCP/IP by Lawrence Robert in April (1967).

3) Server virtualization is a key technology that enables cloud computing. It divides a singles physical server into multiple vms using hypervisor. Each virtual machine its own operation system and application.

Role of server virtualization in cloud.

=> Efficient Resource utilization in cloud.

=> cost Reduction.

=> Rapid provisioning

=> scalability

=> Isolation and security

Virtualization.

=> Enabling technology

=> Creates vm's using hypervisor.

=> process on hardware abstraction and resource

=> Improves serves utilization.

cloud computing.

=> A service delivery model

=> Delivers compting services over the internet

=> Focuses on providing on-demand services to user

=> provides Iaas, Paas, Saas, and Xaas service

Data center:-

It is a facility houses the physical infracture required for cloud computing servers, storage systems networking devices, power supplies and cooling system cloud service providers such as Amazon web services, microsoft and Google cloud operate large data centers to deliver reliable and scalable cloud service to users world wide.

Role of Data centers in cloud services.

=> Computing resource

=> Storage service

=> Networking connectivity

-> Scalability.

-> Reliability and High Availability.

=> Security.

5) Iaas :

=> Infrastructure as a service

=> provides virtual servers, storage and networking

=> manages hardware virtualization and networking

=> manages os, application and data

=> Eg. Amazon EC2 (AWS)

Paas:-

* platform as a service.

* provides platform to develop and deploy application

* manages infrastructure, os, runtime and development tools

* Develops and manages application and data

* Eg : Google App Engine

Saas:-

=> software as a service.

=> provides ready to use software through the internet

=> manages the entire. Application and infrastu-

=> uses the softwre.

=> Eg: Google workspace (Gmail).